



Original Article

Understanding Landowner Intentions to Create Early Successional Forest Habitat in the Northeastern United States

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ABSTRACT Early successional forest habitat (ESH) and associated wildlife species in the northeastern United States are in decline. One way to help create early successional forest conditions is engaging private forest landowners in even-aged forest management because their limited participation may have contributed to declines in ESH for wildlife species of high conservation concern. We applied the reasoned action approach from social psychology to predict intentions of landowners in the 13-county Southern Tier of New York State, USA, to conduct patch-cuts, which is a type of even-aged forest management. We tested the predictive ability of the model using data from a mail survey of landowners conducted from November 2010 to January 2011. Landowner intention to conduct patch-cuts was high (55% of respondents), with attitude being the strongest direct predictor of behavioral intention. Our results suggest that patch-cutting intentions are most likely expressed by landowners who think the behavior is good for their land and wildlife, believe in positive outcomes of land and wildlife management, belong to a game wildlife organization, and have conducted patch-cuts in the past. Strategies to engage more landowners in ESH management will have the highest likelihood of success if outreach efforts focus on influencing behavioral beliefs and subsequently attitudes, possibly working with game wildlife organizations to communicate a unified message for habitat conservation, including the importance of maintaining and creating ESH. Our results demonstrate the importance of social science research to increase the likelihood that conservation targets for declining wildlife species are met. Published 2015. This article is a U.S. Government work and is in the public domain in the USA.

KEY WORDS attitudes, beliefs, early successional habitat, landowners, New York State, reasoned action approach.

Habitat and wildlife species in decline are in immediate need of management. Understanding the correlates of human behavior is essential for successfully implementing programs to encourage actions that contribute to the management and protection of wildlife habitat. In the eastern United States, forest-management behavior of landowners is key to addressing the decline of early successional forest habitat (ESH) and associated wildlife species. Early successional forest habitat, also known as shrubland and young forest, are characterized by persistent shrubs or seedling-sized to sapling-sized trees that are typically a response to some form of natural or human-

induced disturbance (Litvaitis 2003a). Declines in many ESH-reliant taxa have been documented in eastern North America, including plants (Latham 2003), birds (Askins 1993, Dettmers 2003, Rosenberg and Burger 2008), mammals (Litvaitis 1993, 2001; Fuller and DeStefano 2003), and reptiles (Kjoss and Litvaitis 2001). Although declines are due, in part, to changing land-use patterns (i.e., abandonment of agricultural lands in the mid-19th Century), habitat conversion, forest fragmentation, and suppression of natural disturbances have exacerbated declines (Litvaitis 2003a, b). Reversing declines in the ESH-reliant suite of birds has been identified as a conservation priority at the national (e.g., Rich et al. 2004, North American Bird Conservation Initiative 2009), regional (e.g., Dettmers 2003), and state levels (e.g., New York State Department of Environmental Conservation [NYSDEC] 2006). Conservation concern for ESH-reliant species such as golden-winged

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warbler (*Vermivora chrysoptera*), American woodcock (*Scolopax minor*), and New England cottontail (*Sylvilagus transitionalis*) has resulted in major conservation funding initiatives and even proposals for species listing under the U.S. Endangered Species Act.

Forest management is often promoted as a tool to create and maintain sufficient ESH to sustain ESH-reliant wildlife populations (Brooks 2003), especially in the absence of sufficient natural disturbances (e.g., fire, openings from beaver [*Castor canadensis*] activity) and where native shrubland habitat has been converted to other land uses (Litvaitis 2003a). Even-aged timber management (e.g., group selection, clearcutting, seed-tree, and shelterwood) is thought to be one of the most effective means for creating ESH (Costello et al. 2000, King et al. 2001). In addition, because forests are largely (83%) privately owned in the northeastern United States (Miles 2013), forest management for ESH-associated wildlife species relies heavily on the actions of private landowners. Among these private landowners, the amount of timberland owned by nonindustrial forest landowners is 5 times that owned by industrial forest owners (Brooks 2003). New York State, USA, along with other northeastern states, has prioritized activities to increase the amount of ESH as stated in their Comprehensive Wildlife Conservation Strategies and State Wildlife Action Plans (NYSDEC 2006). Despite this priority, at the time that this research was conducted, NYSDEC did not have any programs to support landowner interest in managing for ESH. With an interest in starting such a program, NYSDEC was aware that the effectiveness of programs targeting landowners can be limited by a lack of understanding of how and why private landowners make decisions to manage for ESH (Gobster 2001).

Research concerning people's beliefs, attitudes, and behavior relative to ESH and its management has been very limited. One study reported that residents of the northeastern United States were largely unaware that ESH is declining (Enck and Odato 2008). Another study (Dayer et al. 2014a) found that education programs would be more likely to influence landowner behavior to manage for ESH than financial incentives. Further, tools to engage landowners in management may reinforce behaviors, rather than engage those landowners who are unengaged. Dayer et al. (2014b) assessed typologies for segmenting landowners into groups for targeted outreach related to ESH management.

Gobster (2001) suggested that landowners' willingness to manage for ESH could be predicted from a suite of factors including their interest in trees and nontimber forest products, visual and aesthetic perceptions, attitudes toward clearcutting, and recreational use. Wildlife-related recreation and protection of nature-biological diversity are important reasons landowners cite for owning their woodlands (Connelly et al. 2007), yet many do not conduct forest management to maintain or enhance wildlife habitat and related benefits (Sinclair and Knuth 2000). Landowner attitudes toward clearcutting are particularly relevant to ESH management. Clearcutting is an even-aged cutting approach

that tends to result in ESH during the seedling-sapling regeneration stage. Bliss (2000) suggested that clearcutting has received more criticism than any other forestry practice because of the negative associations of deforestation, plantation forestry, environmental degradation, and aesthetic impacts. However, he found that clearcutting was more acceptable on private land, in smaller patches, and, in some cases, among those with greater ecological knowledge about forest practices, which suggests that private forest landowners could potentially be a viable means of increasing ESH.

We investigated factors that influence private landowner decisions to conduct forest management for ESH. We applied the reasoned action approach (RAA), a social-psychological theory used to understand factors that influence behavioral intention, and ultimately behavior (Fishbein and Ajzen 2010). Our goal was to evaluate factors that influence private landowner intentions to conduct even-aged forest management that can result in ESH, specifically small clearcuts >0.2 ha (hereafter, referred to as patch-cuts). Past studies of landowners have found that concepts from the RAA (i.e., attitude, perceived norm, perceived behavioral control, beliefs) were useful for understanding behavioral intentions to manage land (e.g., Corbett 2002, Karppinen 2005, Vogt et al. 2005). We explored how these concepts can help explain differences in landowners' intentions to create patch-cuts. We also explored the role of the RAA model's "background factors" in influencing beliefs, by developing and empirically testing an adaptation of the RAA model for forest landowners. Specifically, we compared the role of "landowner characteristics" (i.e., membership in game wildlife organizations, residence, past behavior) and "land characteristics" (i.e., total acres owned, acres in mature forest, acres in ESH) in predicting beliefs, attitudes, norms, perceived behavioral control, and behavioral intentions related to ESH management on private lands. Our results provide the first quantitative assessment of landowner behavioral intentions to create patch-cuts, which can guide agencies and organizations in fostering ESH management on private lands.

STUDY AREA

Our study focused on the Southern Tier region of New York State. This region includes the counties of Chautauqua, Cattaraugus, Allegany, Steuben, Schuyler, Chemung, Tompkins, Cortland, Tioga, Broome, Chenango, Otsego, and Delaware. These counties are located on, or near, the border of New York and Pennsylvania, USA. This area was selected based on its mature forest cover compared with other areas in the state, and for the potential benefit for creation of ESH (Dayer et al. 2011).

According to the U.S. Department of Agriculture's Forest Inventory and Analysis (FIA) data from 2007, 7.8% of the forestland area statewide was in the seedling-sapling stage suitable as ESH (Miles 2013). Because particular bird species' habitat needs have been shown to be narrower than the FIA definition for ESH (Schlossberg and King 2009), the availability of ESH suitable for ESH-reliant songbirds in

New York is estimated to be only 3.9% of the forested landscape. This is approximately one-fourth of that believed to be necessary for maintaining bird populations in the region (Dettmers 2003). Of the 789,137 ha of seedling–sapling forestland in New York, 89% is in private ownership (Miles 2013); compared with 77% for all New York State forestlands (Butler 2008).

METHODS

Applying the Reasoned Action Approach to Landowner Behavior

We first developed a conceptual model that explicitly linked landowner and land characteristics to the landowner's intention to conduct ESH management (Fig. 1). In the RAA, one's behavioral intention (or "readiness to perform the behavior") is the best single indicator of behavior (Fishbein and Ajzen 2010). Predictors of intention include attitudes (i.e., tendencies to respond with a degree of favorableness) toward the behavior, perceived norms (i.e., perceptions of what is acceptable behavior in a group or society; hereafter, "norms") to conduct the behavior, and perceived behavioral control (i.e., perceptions of one's control over a behavior; PBC [Fishbein and Ajzen 2010; Fig. 1]).

At the next level of the model, a set of beliefs influences behavioral intention indirectly through each of the direct determinants illustrated in Figure 1. Behavioral beliefs (i.e., perceptions of likely consequences) influence attitudes, normative beliefs (i.e., perceptions of demands placed on humans by others) influence norms, and control beliefs (i.e., perceptions of resources and barriers) influence perceived behavioral control (Fishbein and Ajzen 2010). Fishbein and Ajzen assert that understanding how attitudes, norms, or perceived behavioral control affect behavioral intention is

most important for designing behavior-change campaigns. Finally, background factors are included in the RAA as the origins of these beliefs; these include socio-demographic attributes, culture, exposure to information, emotion, personality, perception of risk, values, and past behavior. The specific relationships between background factors and beliefs and other concepts were not part of Fishbein and Ajzen's (2010) original RAA theory.

To assess the influence of specific background factors on landowner intentions to conduct ESH management, we extended the RAA model to include landowners' past behavior—a landowner characteristic that has commonly been included in applications of this theory to landowner research (Corbett 2002, Karppinen 2005, Vogt et al. 2005). Given that the relationship between past behavior and future behavior is often not fully mediated by predictors in the RAA, researchers have suggested that it be added to the theory (Ouellette and Wood 1998, Albarracín et al. 2001). Further, it has been shown that past behavior can also predict behavioral intentions, even after controlling for its effect on attitudes, norms, and PBC (Terry et al. 1999, Albarracín et al. 2001, Sandberg and Conner 2008, Rise et al. 2010). Meta-analyses have found that adding past behavior to the prediction of behavioral intention increases explained variance by approximately 10% (Sandberg and Conner 2008, Rise et al. 2010).

Additionally, we added other variables that have been found to explain landowners' beliefs and/or management behavioral intentions, including land-ownership variables such as length of land tenure and amount of land owned (e.g., Langpap 2004, Poudyal and Hodges 2009), wildlife-related organizational membership (e.g., Nagubadi et al. 1996, Langpap 2004), and land characteristics such as past

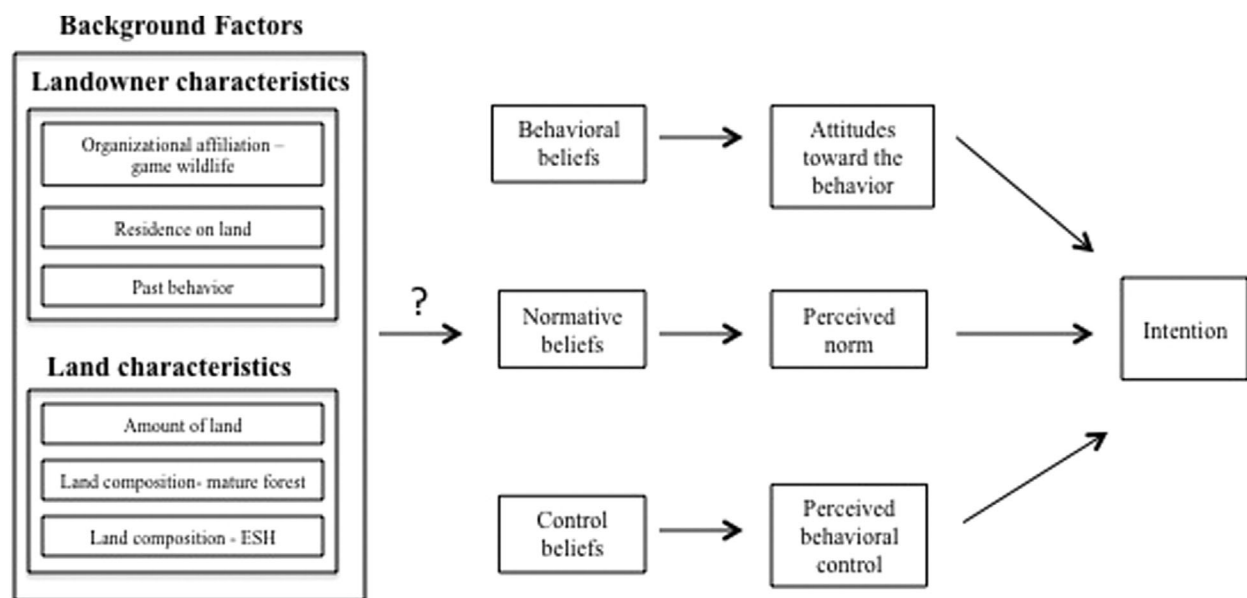


Figure 1. Conceptual model for determining landowner behavioral intentions (based on Fishbein and Ajzen's [2010] reasoned action approach), including predictors of intention, beliefs, and background factors (landowner and land characteristics). The relationship between background factors and the other concepts in the model is not predicted by the theory. ESH is early successional forest habitat.

silvicultural practices of a site (Ross-Davis et al. 2005) and parcel size (Fig. 1).

Survey of New York Southern Tier Landowners

We employed sequential mixed methods (as recommended for landowner studies by Bliss and Martin 1989, Hodgdon et al. 2007). We first interviewed 29 subject matter experts (i.e., professionals who specialize in ESH research, outreach, or management), followed by 32 interviews with landowners experienced in ESH management (Dayer et al. 2011). These qualitative findings informed the design of a quantitative landowner survey. We present only the quantitative data in this article. All phases of our research were reviewed, under protocol 1006001472, by the Cornell University Office of Research Integrity and Assurance and qualified for Exemption from the Institutional Review Board.

Sampling.—We conducted a mail survey of a stratified random sample of landowners who owned ≥ 4 ha of land in ≥ 1 contiguous parcel in 1 of the 13 Southern Tier counties. This parcel size was selected as the minimum given that likelihood of landowners conducting forest management decreases as parcel size decreases (Vickery et al. 2009). Previous studies have found that parcel size influences forest landowner attitudes, behaviors, and intentions (Butler 2008); therefore, we devised a sampling approach so that it would provide an adequate number of responses from landowners with forestlands of various sizes (≥ 20 ha and 4–20 ha). From each sampling frame, we mailed surveys to 1,250 potential respondents (2,500 total landowners).

The sample was drawn from tax code records obtained from the New York Department of Taxation and Finance Office of Real Property Tax Services. Selection of participants was limited to parcels with Office of Real Property tax codes that might include private forest landowners, excluding properties that represented businesses (e.g., Limited Liability Corporations), given our focus on individual or family forest landowners. For a further description of the sampling approach, see Dayer et al. (2011).

Survey design and measurement.—We designed the mail survey instrument as an 8.5 $\times$ 11-inch (21.6 $\times$ 27.9-cm) booklet. It asked landowners to respond by considering all of their parcels of land in the Southern Tier. Kauneckis and York (2009) advocated for asking landowners about their land portfolio (or all of the land owned and managed by a landowner) given that 1) management often transcends a single parcel, and 2) landowners have been found to have difficulty distinguishing individual parcels and most consider parcels as combined except for tax purposes.

The dependent variable in the model (Fig. 1) was landowner behavioral intentions to create a patch-cut, worded in the survey as, “cutting a patch of trees ($\geq$ a $\frac{1}{2}$ acre; 0.2 ha), where all or most trees were removed (to open the canopy) and then plants and trees were allowed to grow back.” The word “clearcut” was not employed given concerns that this term can evoke a negative response. Additionally, respondents could select that they “do not own land with woods” as part of the item on behavioral intention and past behavior. For more information about how variables were

measured, see Table 1 and Dayer (2013). The survey instrument can be found in Dayer et al. (2011).

Data collection.—We collected data from November 2010 to January 2011 using mail-back questionnaires following a modified Tailored Design Method approach (Dillman et al. 2009). The approach consisted of up to 4 mailings and included a cover letter and questionnaire, thank you–reminder postcard, cover letter and replacement questionnaire, and reminder postcard. One to 2 weeks passed between each of the mailings.

In total, 1,036 individuals responded to the survey (521 from the 4–20-ha strata; 515 from the ≥ 20 ha strata). After accounting for undeliverables, the overall response rate was 43%. For the purpose of the model analysis, both landowner strata were combined to ensure adequate sample size to test a model with numerous predictors. It was not possible to weight the sample because many respondents owned >1 parcel, and a population estimate was unavailable for the total amount of land.

A telephone survey was administered to a random sample of 50 nonrespondents from each stratum to identify any sources of nonresponse bias. The Cornell University Survey Research Institute conducted the telephone surveys, which included a subset of items (i.e., land characteristics, preferences for land characteristics, attitudes, ownership motivations, beliefs, cutting behavior, gender, age, and education) from the mail-back survey to allow for a comparison between respondents and nonrespondents to the mail survey. The t -test and χ^2 tests revealed that the respondents and nonrespondents differed significantly ($P < 0.05$) on 2 land characteristics, several motivations, behavioral intentions, and education level (see Dayer 2013). However, the effect sizes of all of these differences were minimal (Vaske 2008), so the respondent sample was not weighted to adjust for nonresponse bias.

Analysis

We conducted our analyses with IBM SPSS Statistics for Windows, Version 21.0 including IBM SPSS Amos 21 (International Business Machines Corporation, Armonk, NY; IBM Corporation 2012). To assess our research questions associated with the RAA with a dichotomous-dependent variable (i.e., behavioral intention), we employed Bayesian structural equation modeling using Markov chain Monte Carlo simulations. Structural Equation Models (SEM) allow for evaluating a series of simultaneous hypotheses about impacts of variables on each other, while also taking measurement error into consideration (Lee 2007). The Bayesian SEM (BSEM) approach is appropriate for models with ordered categorical variables, dichotomous variables, and missing data (Lee 2007, Song and Lee 2012).

We constructed a sequential series of BSEMs, starting with the most simple attitude–norm–PBC model and adding levels of complexity in each successive model, following the steps in our conceptual model (Fig. 1), to test whether these added components improved our ability to predict the behavioral intention of landowners to cut patches of trees. To evaluate the added effects of beliefs on predictors of

Table 1. Means and standard deviations for variables, resulting from a 2010 mail survey of private forest landowners in 13 counties of the Southern Tier of New York, USA.

Concepts	Mean	SD
Behavioral intention ^a	0.55	0.50
Attitudes toward the behavior ^{b,c}	3.40	1.04
Perceived norm ^d	2.18	1.03
Perceived behavioral control ^e	3.09	1.03
Behavioral belief ^{f,g}	3.64	0.73
Normative belief ^{c,h}	3.02	0.88
Control belief ⁱ	2.68	0.68
Organizational affiliation—membership in game wildlife organization ^j	0.04	0.14
Residence on land ^k	0.61	0.49
Past behavior ^l	0.33	0.47
Amount of land (ha) ^m	51.79	63.89
Land composition—mature forest (%) ⁿ	0.41	0.30
Land composition—early successional habitat (% shrubland and young forest) ^o	0.21	0.32

^a Response to mail survey item “cutting a patch of trees ($\geq$ a $\frac{1}{2}$ acre; 0.2 ha) where all or most of the trees were removed (to open the canopy) and then plants and trees were allowed to grow back.” Collapsed from a categorical variable into a binary variable with 0 for no intention to cut in the next 5 years and 1 for “slightly,” “somewhat,” or “very” likely to cut in the next 5 years.

^b Mean scale of response to 2 Likert-type items—1 on the perceived result for “your land” and 1 for “wildlife.”

^c Items were assessed on a 5-point scale: 1 = “very bad,” 2 = “bad,” 3 = “neither,” 4 = “good,” 5 = “very good.”

^d Response to survey item “how common it is that other landowners in your areas do this activity.” Item assessed on a 4-point scale: 1 = “not at all common” to 4 = “very common.”

^e Response to survey item “to what extent do you feel you are able to get the following activity done on your land [if you decide to do so]?” Item assessed on a 4-point scale: 1 = “not at all able” to 4 = “very able.”

^f Composite mean scale of 6 belief items about benefits of management of land and wildlife (see Dayer 2013 for items). The Cronbach’s α for the scale was 0.83.

^g Items were assessed on a 5-point scale: 1 = “strongly disagree,” 2 = “disagree,” 3 = “neither,” 4 = “agree,” 5 = “strongly agree.”

^h Response to survey item “How bad or good do nearby landowners think cutting a patch of trees ($\geq$ $\frac{1}{2}$ acre) where all or most of the trees were removed would be for your land?”

ⁱ Composite mean scale of 11 belief items about potential barriers to performing patch-cuts (see Dayer 2013 for items). The Cronbach’s α for the scale was 0.86.

^j Composite mean scale for survey item “which wildlife or land organizations are you a member of?” with options National Wild Turkey Federation, Pheasants Forever, Quality Deer Management Association, and Ruffed Grouse Society. We considered game-wildlife organization membership, rather than membership in other types of organizations, based upon our interview research in the formative phase of this study (see Dayer et al. 2011). The scale ranged from 0 (membership in no organizations) to 1 (membership in 3 organizations).

^k Measured as “yes” (if distance from any of the parcels was <1 mile) or “no” (if distance from all parcels was >1 mile).

^l Response to survey item “Have you done this activity in the last 10 years?” with the same patch-cutting behavior description as for behavioral intention. Item was dichotomous: 1 = “yes,” 0 = “no.”

^m Sum of acres on all parcels in the Southern Tier.

ⁿ Calculation of percentage of acres reported in mature forest divided by total acreage in all habitat types. See description and image of habitat types in Dayer et al. (2011).

^o Calculation of percentage of acres reported in shrubland and young forest divided by total acreage in all habitat types.

intention, we compared the attitude–norm–PBC BSEM to a model that added behavioral beliefs predicting attitudes; normative beliefs predicting perceived norm; and control beliefs predicting perceived behavioral control. To test whether the influences of beliefs on behavioral intention were mediated through the predictors of intention, we additionally explored the relationship between beliefs and behavioral intention, controlling for the direct determinants. To assess whether land or landowner characteristics as background factors provided an improvement to the RAA for understanding behavioral intentions to patch-cut, we compared models with land or landowner characteristics added to the model without these variables. In each model, we considered the relationship between each characteristic and each belief separately. Additionally, we assessed the relationship between past behavior and both the predictors of intention and behavioral belief for the land characteristic model and the landowner characteristic model.

For BSEM analyses, models are considered to have converged when the convergence statistic falls below 1.002 (Arbuckle 2009). For model comparisons, we used the

goodness-of-fit measures of posterior predictive P -value (PPP) and the deviance information criterion (DIC). The PPP assesses the model’s specification quality in terms of predictive accuracy (Kaplan and Depaoli 2012). A PPP approximately 0.50 indicates a plausible model, whereas one closer to 0 or 1 indicates that the model is not plausible (Song and Lee 2012). When comparing models, the model with the smallest DIC is preferred (Kaplan and Depaoli 2012). Posterior means, which are the average of the posterior distribution, were reported as the parameter estimate (Arbuckle 2009). Significance of individual posterior means was examined via credible intervals (i.e., the Bayesian equivalent of CIs). Posterior means were determined to be significant when the 95% credible intervals (also known as posterior probability interval) excluded zero.

RESULTS

Characteristics of Forest Landowners

Landowners responding to our survey were primarily male (82%) and averaged 61 years old ($SD = 12$). Thirty-eight

percent had a 4-year college undergraduate (Bachelors) or higher degree. Sixty-one percent lived on one of their land parcels in the Southern Tier. Landowners owned their land for an average of 23 years ($SD = 16$), and owned a mean of 52 ha ($SD = 64$) across all of their parcels in the Southern Tier. Thirteen respondents owned no land with forest cover and were eliminated from the model analyses. For means and standard deviations of all variables, see Table 1.

Predictors of Behavioral Intention

The simplest Bayesian SEM that evaluated the effects of attitudes, norm, and PBC on behavioral intention (Model 1; Table 2) suggests attitudes toward land and wildlife positively predicted the landowner intentions to conduct patch-cutting in the next 5 years ($\beta = 0.24$). However, norms related to the behavior and perceived behavioral control were not significant predictors of behavioral intention. The PPP (0.50) indicates a good model fit, and the DIC was 4,177.43 (Table 2).

Effect of Beliefs on Predictors of Intention

When beliefs were added to the simpler model, we found that beliefs (i.e., behavioral, normative, control) each positively predicted their respective predictors of intention (Model 2; Table 2). The relationship of behavioral beliefs with attitudes was the strongest ($\beta = 0.70$), followed by control beliefs with perceived behavioral control ($\beta = 0.57$), and then normative beliefs with a perceived norm ($\beta = 0.36$). None of the beliefs directly predicted intention. When compared with the model with just the predictors of intention, the goodness-of-fit (DIC = 4,829.67) was higher, although the PPP was still adequate (0.52).

Effects of Landowner and Land Background Characteristics

When comparing the land background characteristics (Model 3) and landowner background characteristics (Model 4) models, the landowner characteristics model performed better (DIC = 801.17) than the land characteristics model (DIC = 15,393.70; Table 2). Both models had adequate measures of PPP (0.52). In the land model, only the percentage of mature forest significantly and positively predicted control beliefs ($\beta = 0.20$). Otherwise, land

characteristics had no relationship with beliefs. All landowner characteristics significantly predicted beliefs: positive behavioral beliefs were associated with membership in a game wildlife organization ($\beta = 1.02$) and past behavior of patch-cutting ($\beta = 0.26$), but they were not associated with residence on the land. More positive normative beliefs were associated with not living on one's land ($\beta = -0.21$) and past behavior ($\beta = 0.48$), but they were not associated with membership in a game wildlife organization. More positive control beliefs were associated with membership in a game wildlife organization ($\beta = 0.44$), residence on land ($\beta = 0.14$), and past behavior ($\beta = 0.23$).

Further exploration of the relationship between the background factors (i.e., land and landowner characteristics) and predictors of intention and behavioral intention produced a model with improved fit for the landowner characteristics model (Model 6; Table 2; DIC = 545.99; PPP = 0.51), but no change for the land characteristics model (Model 5; DIC = 15,392.58; PPP = 0.50). Land characteristics did not have a direct effect on intention or its predictors beyond that explained by beliefs. In contrast, all 3 landowner characteristics had significant effects. Past behavior had a direct, independent, and positive relationship with each variable: behavioral intention ($\beta = 0.19$), attitudes ($\beta = 0.82$), perceived norm ($\beta = 0.32$), and PBC ($\beta = 0.32$). The other 2 landowner characteristics—membership in a game wildlife organization ($\beta = 0.60$) and residence on land ($\beta = -0.23$)—directly affected attitudes, beyond that explained by their relationship with beliefs. Of all the models we tested, the most complex was Model 6, which included the predictors of behavioral intention, the added effects of beliefs on those predictors, and inclusion of landowner background characteristics; it had the lowest DIC and was considered the best overall model (Fig. 2).

DISCUSSION

Our results suggest that education, outreach, and communications to engage landowners in even-aged forest management for ESH conservation should focus on behavioral beliefs and attitudes related to managing forests and wildlife and the positive outcomes of even-aged

Table 2. Goodness-of-fit measures and predicted relationships between variables (predictors of intention, beliefs, land characteristics, and landowner characteristics) for 6 Bayesian Structural Equation Models to predict the behavioral intention of private forest landowners to conduct patch-cuts reported in a 2010 mail survey of landowners in the 13 counties of the Southern Tier of New York, USA. Each of 6 successive models tested whether adding additional variables improved significantly on previous simpler models. (For details for each of the models, see Dayer 2013). An X indicates the relationship between the 2 types of concepts was included in a given model.

Predicted relationships	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Predictors of intention → behavioral intention	X	X	X	X	X	X
Beliefs → predictors		X	X	X	X	X
Land characteristics → beliefs			X		X	
Landowner characteristics → beliefs				X		X
Land characteristics → predictors of intention and intention					X	
Landowner characteristics → predictors of intention and intention						X
Goodness-of-fit measures						
Posterior predictive <i>P</i> -value	0.50	0.52	0.52	0.52	0.50	0.51
Deviance information criterion	4,177.43	4,829.67	15,393.70	801.17	15,392.58	545.99

“→” indicates “effect on.”

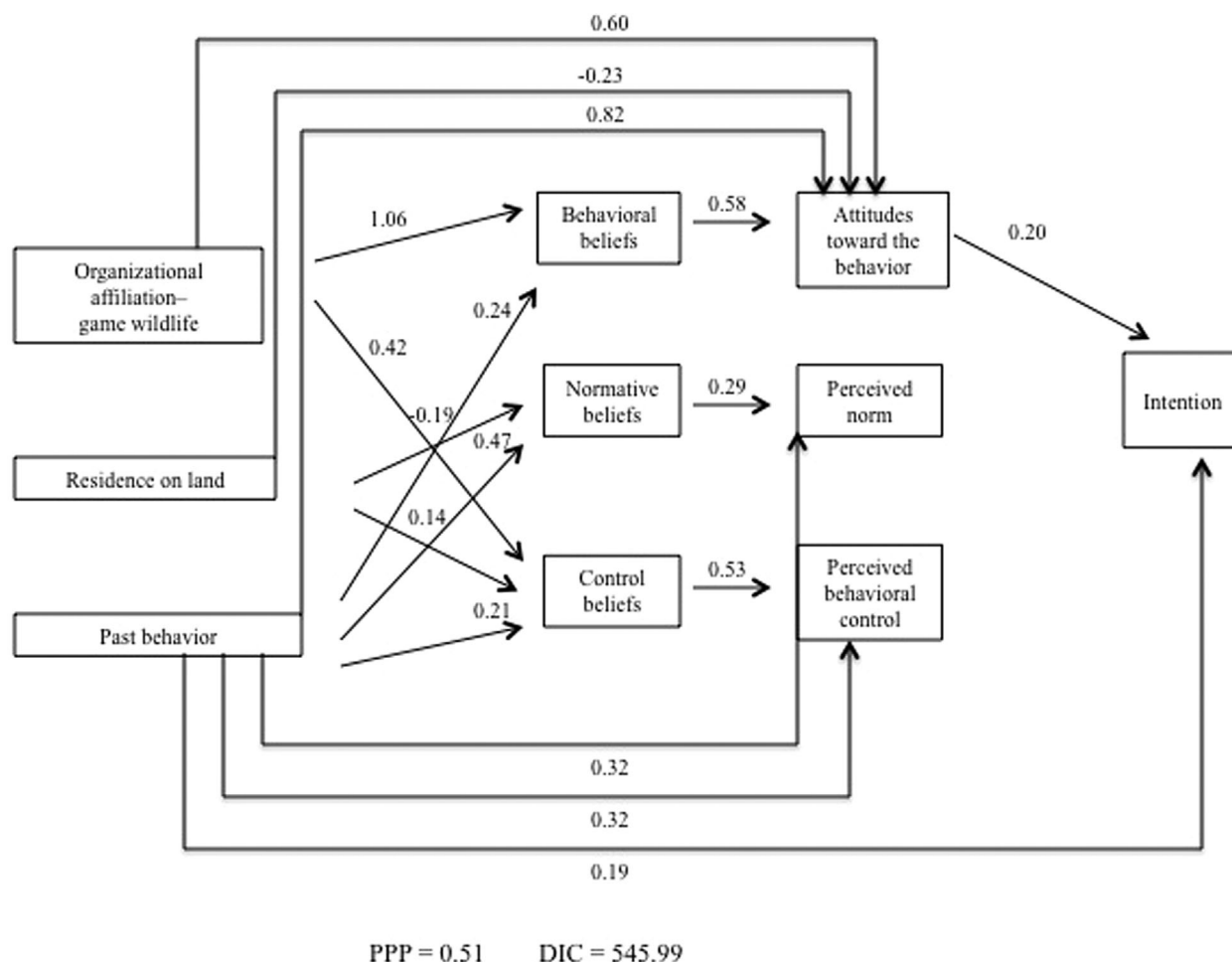


Figure 2. The best fit model (Model 6) for predicting behavioral intentions, including landowner characteristics, beliefs, and predictors of intention, resulting from a 2010 mail survey of private forest landowners in thirteen counties the Southern Tier of New York, USA. Posterior means and pathways are provided for all significant relationships (when the 95% Bayesian credible interval excluded zero). PPP, posterior predictive *P*-value; DIC, deviance information criterion.

management for wildlife and land. Among the predictors of intention under the RAA, landowner attitudes toward patch-cutting—specifically, landowners who thought cutting patches of trees was good for their land and good for wildlife—most strongly influenced behavioral intention to conduct patch-cuts. Somewhat counter to RAA theory, neither perceived behavioral control, nor perceived norm, affected landowners' behavioral intention. This finding is consistent with other landowner research where attitudes were the strongest significant predictor of landowner behavioral intentions (Corbett 2002, Artikov et al. 2006, Thompson 2010). Landowner beliefs significantly predicted attitudes, norms, and PBC. Additionally, behavioral belief indirectly influenced behavioral intention through attitude. This finding reinforces the importance of including underlying beliefs in behavioral models to better understand landowner behavior; past research rarely included beliefs, but rather relied on simpler models with fewer RAA variables.

The addition of landowner characteristics greatly improved our understanding of landowner beliefs. Most important to consider are the landowner characteristics that influence behavioral beliefs and attitudes, given that these variables

were the only significant predictors of behavioral intention. Fishbein and Ajzen (2010) concluded that because background factors tend to have an indirect, mediated relationship with behavioral intention, the best strategy is to not focus on the background factors in application of the model results. Unlike Fishbein and Ajzen (2010), however, we found that landowner characteristics influenced attitude directly; therefore, we recommend that those interested in influencing behavioral intentions pay closer attention to landowner characteristics.

Among landowner characteristics, past cutting behavior was the only variable that influenced all 3 beliefs and also had a residual effect on all other model variables, beyond that influenced through beliefs. Given that the predictors in the RAA do not fully mediate the relationship between past behavior and behavioral intention (Ouellette and Wood 1998, Albarracín et al. 2001), researchers have suggested adding past behavior to the theory. Rather than a result of habituation as some believe it to be (e.g., Ouellette and Wood 1998, Verplanken et al. 1998), Fishbein and Ajzen (2010) suggest that this residual effect of past behavior on intentions could be the result of missing components of the

model (e.g., self-identity and anticipated effect) that explain variance not explained by the theory's main predictors. We suggest that future research explore the role of these additional factors, such as self-identity and anticipated effect, to increase our understanding of landowner behavior.

In contrast to landowner characteristics, land characteristics played little to no role in understanding beliefs. This finding runs counter to other research that emphasizes the role of parcel size in landowner behavior research (Beach et al. 2005, Kauneckis and York 2009). This finding, however, corresponds with recent research on harvest behavior in Pennsylvania that found social factors played a greater role in predicting behavior than do land characteristics (Metcalf 2010). Given that we measured self-reported land characteristics, we suggest that future research still consider actual land characteristics (such as habitat cover and density of wildlife indicator species).

Our human dimensions research approach provides an enhanced understanding of landowner behavior that can serve as guidance for how to improve efforts to engage landowners in cutting patches of trees to create ESH. Adapting an approach used by Playa Lakes Joint Venture to link biological planning data with landowner survey findings (Bogart et al. 2009), we assessed the extent to which following the guidance from this research might have a population-level impact on ESH-reliant birds if applied at a landscape scale. Based on estimates of forest cover and landownership patterns in the 13 Southern Tier counties of New York (Butler 2008, Miles 2013), we calculated that 1 million ha was available for ESH management by our study population of roughly 19,860 private forest landowners. Based on our survey result that 55% of landowners indicated at least a "slight" intention to conduct a patch-cut within the next 5 years, and assuming an average patch-cut of 2 ha, we further estimated that 22,102 ha could potentially be treated with patch-cuts in the next 5 years—or roughly 2% of forested lands owned by individual private landowners.

Using Dettmers' (2003) estimate that 10–15% of the forested landscape should be in ESH to support viable regional populations of ESH-reliant birds (with 1.3–1.9% of the landscape converted to ESH every 5 years), our study population of private forest landowners could meaningfully contribute to this target—assuming other types of landowners (e.g., industrial and public lands) also contribute. Increasing education, outreach, and communications to a targeted group of landowners (i.e., landowners who are members of game wildlife organizations, those not residing on their land, and those who conducted the behavior in the past), as suggested in this study, would increase the likelihood that landowners with an intention to cut patches would act on those intentions. Furthermore, expanding the target audience of landowners through additional outreach, coupled with long-term monitoring of landscape and bird-population changes, should help to reverse declines of ESH-reliant species in the northeastern United States.

Engaging private landowners in habitat conservation will help to meet the goals of landscape-level wildlife conservation. Recently, nongame conservation initiatives such as the

Golden-winged Warbler Working Group (2013; <http://www.gwwa.org>) have partnered with more traditional game wildlife organizations such as the Wildlife Management Institute in delivering outreach tactics and tools and sharing similar messages, and in developing management prescriptions that benefit high-priority nongame species as well as American woodcock and white-tailed deer (Bakermans et al. 2011). As a result of this unified messaging, major funding initiatives have emerged (e.g., U.S. Department of Agriculture Natural Resources Conservation Service, National Fish and Wildlife Foundation) to engage private and public land managers in creating tens of thousands of acres of new ESH across the eastern United States. The success of these initiatives in affecting forested landscapes at sufficient scale will depend on an improved understanding of behavioral beliefs and attitudes of all forest landowners in order to expand the target audiences beyond traditional game-wildlife enthusiasts.

This study offers the first application of the RAA to landowner research, providing insights into how conservation initiatives, agencies, and organizations that typically focus on the biological components of conservation planning and implementation can better understand landowner behavior, identify how landowners vary, and develop tools to better engage landowners with an interest in conservation activities on their land. Better incorporating information from research like ours on the human dimensions of private lands conservation into program design will be critical as conservation efforts seek to protect existing habitat on private lands, and implement management activities to improve habitat quality and quantity, as well as conserve the wildlife species that people value.

MANAGEMENT IMPLICATIONS

Our findings in this study and elsewhere suggest a shift away from the current emphasis on landowner financial incentives as the primary—if not exclusive—approach for encouraging early successional habitat management. Based on our results, it seems unlikely that merely providing financial support for management activities will significantly influence landowner management for ESH. Instead, this research suggests that inadequate participation in these incentive programs may be due to lack of understanding by landowners regarding the value of creating ESH. Focusing on landowner outreach as an initial first step, we believe could create demand for financial incentives for ESH. We suggest that this outreach (e.g., print materials, educational workshops) employ messages emphasizing how patch-cuts (and ESH management generally) can benefit the land and wildlife. Considering landowner characteristics may help in locating landowners who already have an intention to manage for ESH (i.e., landowners who are members of game wildlife organizations, those not residing on their land, and those who conducted the behavior in the past). Additionally, because past behavior played such a strong role in our model, experienced landowners should be specifically engaged through peer support programs or hosting demonstration site or field visits. These activities can both reinforce their

own management practices and encourage others to follow suit. Our past research presents typologies for segmenting landowners into groups with similar characteristics for outreach related to ESH management. We also recommend building on the education and outreach approaches already being implemented by game organizations, such as landowner workshops, demonstration areas, best management practices documents, videos (e.g., <http://www.youngforest.org/content/young-forest-video>), and websites (e.g., timberdoodle.org). Although we cannot be sure whether the relationship between game wildlife organization membership and behavioral belief and attitude is causal, our landowner interview results and an informal review of game wildlife organization outreach materials (e.g., e-newsletters, magazines, websites, sponsored television shows) highlight a wealth of habitat conservation messaging being delivered by these groups.

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